Finite-depth Quasi-Phase Matching (QPM) and Converter Efficiency

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Consider an ensemble of ^{85}Rb atoms coupled to an optical cavity near 780 nm and a microwave cavity. A blue control field near 480/481 nm connects the optically excited and Rydberg states, while UV dressing near 297 nm enables an effective coupling between the prepared atomic state and the microwave-addressed Rydberg excitation. These interactions form the conversion loop considered in Ref. [2]. An additional 1188 nm lattice confines the atoms and supplies a periodic density pattern for quasi-phase matching; it is not one of the resonant conversion fields.

Finite lattice depth leaves each atom distributed over a finite region of a lattice site. The resulting phase spread reduces the coherent overlap, Q, between the atomic modes addressed by the two conversion paths. This note defines that overlap, derives its effect on the optical-to-microwave response, and evaluates a numerical example including cavity input and output coupling. All equations and conventions needed for the derivation are given here. The companion Mathematica notebook implements and checks them; its notation is mapped to this text at the end [3].

1 What the finite-depth QPM factor means

Label the N atoms by $j = 1, \ldots, N$. Let u_j and v_j be the complex excitation amplitudes on atom j in the normalized collective profiles addressed by the optical/blue and UV/microwave paths. Each profile includes the relative coupling magnitudes and optical phases; its overall coupling strength is treated separately in section 2. Define

$$Q = \left| \sum_j u_j^* v_j \right|, \qquad \sum_j |u_j|^2 = \sum_j |v_j|^2 = 1, \qquad 0 \le Q \le 1. \tag{1}$$

For matched, uniform coupling envelopes, this becomes $Q = |\langle e^{i\Delta\mathbf{k}\cdot\mathbf{r}} \rangle|$, where the average is over the normalized atomic spatial distribution. For the chosen conversion loop, $\Delta\mathbf{k} = \mathbf{k}_{780} + \mathbf{k}_{480} - \mathbf{k}_{\mathrm{UV}} - \mathbf{k}_{\mu}$; each wavevector includes its propagation direction. Reversing the loop changes the sign of this mismatch but not the magnitude Q. Thus Q is an *amplitude overlap*; Q^2 is the squared projection onto the useful collective mode. Neither quantity alone is the converter's success probability. The efficiency also depends on coherent exchange, atomic decay, and the cavity ports.

1.1 Localization within one site.

An ideal retroreflected lattice has period $d = \lambda_{\mathrm{L}}/2$ and reciprocal wavevector $K = 2k_{\mathrm{L}}$, with $k_{\mathrm{L}} = 2\pi/\lambda_{\mathrm{L}}$. Let x measure displacement from a site minimum and U be the energy from that minimum to the barrier. For atomic mass m, the recoil energy E_{R} and the harmonic trap angular frequency ω_t are [1]

$$V(x) = U \sin^2(k_{\mathrm{L}} x) \simeq U k_{\mathrm{L}}^2 x^2, \quad E_{\mathrm{R}} = \frac{\hbar^2 k_{\mathrm{L}}^2}{2m}, \quad \omega_t = \frac{2\sqrt{U E_{\mathrm{R}}}}{\hbar}. \tag{2}$$

Here $\hbar$ is the reduced Planck constant. At temperature T, with Boltzmann constant k_{B}, the thermal position variance and the nth density Fourier harmonic are then

$$\sigma_x^2 = \frac{\hbar}{2m\omega_t}\coth\!\left(\frac{\hbar\omega_t}{2k_{\mathrm{B}}T}\right),$$
$$b_n = \left|\left\langle e^{inKx}\right\rangle\right| = e^{-(nK)^2\sigma_x^2/2} = \exp\!\left[-n^2\sqrt{\frac{E_{\mathrm{R}}}{U}}\coth\!\left(\frac{\sqrt{UE_{\mathrm{R}}}}{k_{\mathrm{B}}T}\right)\right]. \tag{3}$$

The integer n labels a spatial Fourier harmonic, not a vibrational band. At $T = 0$, the hyperbolic cotangent tends to one: zero-point motion still gives $b_n < 1$ for any finite U and nonzero n. At fixed temperature, deeper trapping increases b_n toward one. The approximation requires localization near the well minimum; appreciable occupation near the barrier calls for actual band wavefunctions.

1.2 From one axis to the total overlap.

Write an atom's position as its lattice-site centre $\mathbf{R}$ plus its displacement within the site. If $\Delta\mathbf{k} = \sum_i n_i\mathbf{K}_i$ for integer orders n_i, the phase at every equivalent site centre is the same. This is quasi-phase matching: the lattice removes cancellation between sites, while finite localization within each site still reduces the overlap. For independent confined directions, the remaining factors multiply:

$$Q = \prod_i b_{n_i,i}; \qquad Q = b_{2,x}b_{2,y} \quad \text{for two second-order QPM axes.} \tag{4}$$

The last expression is the working geometry used below: co-propagating 780 and 480/481 nm fields, perpendicular UV, and suitable lattice axes. It is a *conditional geometry*, not fixed by the nominal wavelengths alone. A residual mismatch $\delta\mathbf{k}$ also produces phase cancellation across occupied site centres, through $|\langle e^{i\delta\mathbf{k}\cdot\mathbf{R}}\rangle|$. Beam angles, exact wavelengths, and mode envelopes must therefore be checked separately.

2 Analytical converter model with the orthogonal modes

2.1 Problem definition and collective-mode construction.

The system contains an optical cavity mode $\hat{a}$, a microwave cavity mode $\hat{b}$, and an ensemble with optically excited and Rydberg excitations. The optical cavity couples to one collective atomic profile with strength g_o; the UV-dressed microwave interaction couples to another with strength g_m. The blue field connects the optical and Rydberg excitations with matrix element Ω. The profiles need not coincide: their normalized overlap is Q from Eq. (1).

We drive only the optical input and seek the coherent microwave output in the linear, weak-excitation regime. The conversion efficiency is the ratio of output microwave photon flux to input optical photon flux. It is not an energy-efficiency ratio: the control fields account for the difference between the photon energies. All couplings, detunings, and linewidths in the analytical equations are angular frequencies; numerical values divided by 2π are stated explicitly in section 3.

2.1.1 Atomic variables and assumptions.

Let $\hat{e}_j^\dagger$ and $\hat{r}_j^\dagger$ create an optical and a Rydberg excitation, respectively, on atom j relative to the prepared reference state. The optical excitation is a ground-to-optically-excited-state coherence; the Rydberg excitation is the effective coherence addressed by the blue and UV-dressed microwave paths. When the number of

signal excitations is small compared with the number of prepared atoms, depletion and saturation can be neglected and these operators are approximately bosonic:

$$[\hat{e}_j, \hat{e}_\ell^\dagger] \simeq [\hat{r}_j, \hat{r}_\ell^\dagger] \simeq \delta_{j\ell}, \qquad [\hat{e}_j, \hat{r}_\ell^\dagger] \simeq 0. \tag{5}$$

Assume frozen atomic positions during conversion, a uniform blue-control magnitude, and identical atomic detunings and independent decay rates across the ensemble. UV dressing is already included in the effective microwave coupling and its spatial profile. After absorbing the blue field's spatial phase into $\hat{r}_j$, its matrix element Ω can be chosen real and uniform. It is defined as the Hamiltonian matrix element below, so a Rabi-frequency convention that puts a factor of $1/2$ in the Hamiltonian must be converted before substituting a value for Ω. With the sign convention used below,

$$\frac{\hat{H}_B}{\hbar} = -\Omega \sum_j (\hat{e}_j^\dagger \hat{r}_j + \hat{r}_j^\dagger \hat{e}_j). \tag{6}$$

2.1.2 The two coupling profiles.

In this phase convention, let $\mathbf{u}$ describe the collective optical excitation selected by the optical cavity, and $\mathbf{v}$ the Rydberg excitation selected by the UV-dressed microwave interaction. These are the profiles introduced in section 1:

$$\mathbf{u}^\dagger \mathbf{u} = \mathbf{v}^\dagger \mathbf{v} = 1, \qquad \mathcal{Q} = \mathbf{u}^\dagger \mathbf{v} = \sum_j u_j^* v_j, \qquad Q = |\mathcal{Q}|. \tag{7}$$

A global phase of $\mathbf{v}$ can be absorbed into the microwave cavity operator, so we henceforth choose $\mathbf{u}^\dagger \mathbf{v} = Q \geq 0$. Writing $s = \sqrt{1 - Q^2}$, the microwave profile has the exact decomposition

$$\boxed{\mathbf{v} = Q\mathbf{u} + s\mathbf{w},} \qquad \mathbf{w} = \frac{\mathbf{v} - Q\mathbf{u}}{s}, \qquad \mathbf{u}^\dagger \mathbf{w} = 0, \quad \mathbf{w}^\dagger \mathbf{w} = 1. \tag{8}$$

The expression for $\mathbf{w}$ applies when $Q < 1$. At $Q = 1$, choose any normalized orthogonal profile: its coupling to the microwave cavity vanishes because $s = 0$.

2.1.3 Collective operators and closure.

Define four atomic operators, using uppercase letters to distinguish the operators from the coherent amplitudes used later:

$$\begin{aligned} \hat{P}^\dagger &= \sum_j u_j \hat{e}_j^\dagger, \quad \hat{R}^\dagger = \sum_j u_j \hat{r}_j^\dagger, \\ \hat{P}_\perp^\dagger &= \sum_j w_j \hat{e}_j^\dagger, \quad \hat{R}_\perp^\dagger = \sum_j w_j \hat{r}_j^\dagger. \end{aligned} \tag{9}$$

To the same weak-excitation approximation, orthonormality gives unit commutators within each mode; distinct profiles commute. The corresponding coupling chain is

$$a \overset{g_o}{\longleftrightarrow} E_1 \overset{\Omega_B}{\longleftrightarrow} R_1 \overset{Qg_m}{\longleftrightarrow} b \overset{sg_m}{\longleftrightarrow} R_2 \overset{\Omega_B}{\longleftrightarrow} E_2. \tag{10}$$

Here a, b label the optical and microwave cavity modes. The chain labels E_1, R_1, R_2, E_2 denote the modes represented by $\hat{P}, \hat{R}, \hat{R}_\perp, \hat{P}_\perp$, respectively, and $\Omega_B = \Omega$; $s = \sqrt{1 - Q^2}$ as above. The two cavity profiles span at most a two-dimensional atomic subspace, preserved by the uniform control, detunings, and damping. Hence the coherently driven response closes on these four atomic modes plus the two cavity modes. All other orthogonal atomic modes have zero mean drive and are taken initially unexcited.

2.1.4 Hamiltonian and conventions.

The strengths g_o and g_m are collective Hamiltonian matrix elements, including the total coupling magnitude of each normalized profile. They are not single-atom couplings and are held fixed when varying Q. To use the parameter convention of the baseline converter model [3], define

$$g_o = \frac{G}{\sqrt{2}}, \qquad g_m = \frac{G_\mu}{\sqrt{2}}. \tag{11}$$

Here G and G_μ are alternative names for the collective coupling scales, not additional interactions. The parameters Δ_e, Δ_r, and Δ_μ describe the optical atomic, effective Rydberg, and microwave cavity detunings in the rotating frame. Their signs are fixed by the following free Hamiltonian, with zero optical-cavity detuning. Restricted to the driven subspace,

$$\frac{\hat{H}_0}{\hbar} = -\Delta_\mu \hat{b}^\dagger \hat{b} - \Delta_e(\hat{P}^\dagger \hat{P} + \hat{P}_\perp^\dagger \hat{P}_\perp) - \Delta_r(\hat{R}^\dagger \hat{R} + \hat{R}_\perp^\dagger \hat{R}_\perp), \tag{12}$$

$$\frac{\hat{H}_{\rm int}}{\hbar} = -\Big[g_o \hat{a}^\dagger \hat{P} + \Omega(\hat{P}^\dagger \hat{R} + \hat{P}_\perp^\dagger \hat{R}_\perp) + g_m \hat{b}^\dagger (Q\hat{R} + s\hat{R}_\perp) + \text{h.c.}\Big]. \tag{13}$$

The minus signs fix the detuning and coupling-phase convention to match Eq. (16): use $\dot{\hat{c}} = (i/\hbar)[\hat{H}, \hat{c}]$ and Fourier amplitudes proportional to $e^{+i\omega t}$, where ω is the spectral offset from the chosen reference. Let $\kappa_{o,\rm ex}$ and $\kappa_{m,\rm ex}$ be the optical input-port and useful microwave output-port coupling rates, respectively. In this convention an optical drive with photon-flux amplitude $a_{\rm in}(t)$ is represented by

$$\frac{\hat{H}_{\rm d}}{\hbar} = -\sqrt{\kappa_{o,\rm ex}}\,\big[\hat{a}^\dagger a_{\rm in}(t) + a_{\rm in}^*(t)\hat{a}\big]. \tag{14}$$

Here $|a_{\rm in}|^2$ is incident photon flux. Define $a = \langle \hat{a} \rangle$, $b = \langle \hat{b} \rangle$, $p = \langle \hat{P} \rangle$, $r = \langle \hat{R} \rangle$, $p_\perp = \langle \hat{P}_\perp \rangle$, and $r_\perp = \langle \hat{R}_\perp \rangle$. With no incident microwave field, the useful output has amplitude $b_{\rm out} = \sqrt{\kappa_{m,\rm ex}}\, b$ up to an overall port phase, so $S_{mo} = b_{\rm out}/a_{\rm in}$. Reservoir noise has zero mean in this response calculation; the transfer efficiency does not characterize added noise.

2.2 Response functions and linewidth conventions.

Let κ_o and κ_m be the total cavity energy decay rates, including all cavity ports and internal loss. Thus $\kappa_{o,\rm ex} \le \kappa_o$ and $\kappa_{m,\rm ex} \le \kappa_m$. Define the atomic linewidth parameters Γ and γ_r so that the optical and Rydberg amplitudes decay at $\Gamma/2$ and $\gamma_r/2$, respectively.

To preserve the baseline converter equations [3], this model also assigns an additional amplitude damping $\gamma_r/2$ to the microwave mode: its effective full width is $\kappa_m + \gamma_r$. This is a separate phenomenological assumption of the baseline model, not a consequence of QPM or a change to the useful port rate. Including these damping terms, the inverse response functions are

$$\begin{aligned} D_a &= \kappa_o/2 + i\omega, & D_e &= \Gamma/2 + i(\omega - \Delta_e), \\ D_r &= \gamma_r/2 + i(\omega - \Delta_r), & D_b &= (\kappa_m + \gamma_r)/2 + i(\omega - \Delta_\mu). \end{aligned} \tag{15}$$

The real parts set amplitude damping and the imaginary parts describe detuning. The additional response due to the orthogonal atomic modes will be derived separately; it is not included in these definitions.

2.3 Six coupled equations.

The coherent amplitudes defined above obey, with the chosen input phase,

$$\begin{aligned} D_a a - i g_o p &= i\sqrt{\kappa_{o,\mathrm{ex}}}\, a_{\mathrm{in}}, \\ D_e p - i g_o a - i\Omega r &= 0, \\ D_r r - i\Omega p - i g_m Q b &= 0, \\ D_b b - i g_m (Q r + \sqrt{1-Q^2}\, r_\perp) &= 0, \\ D_r r_\perp - i\Omega p_\perp - i g_m \sqrt{1-Q^2}\, b &= 0, \\ D_e p_\perp - i\Omega r_\perp &= 0. \end{aligned} \tag{16}$$

The microwave mode couples to $Qr + \sqrt{1-Q^2}\, r_\perp$. Its total collective coupling remains g_m; the useful component is Qg_m.

2.4 Eliminating the orthogonal pair.

The last equation gives $p_\perp = i\Omega r_\perp / D_e$. Substituting this into the $r_\perp$ equation and writing $F = D_e D_r + \Omega^2$ gives

$$r_\perp = \frac{i g_m \sqrt{1-Q^2}\, D_e}{F}\, b, \qquad \boxed{\Sigma_\perp(\omega) = \frac{g_m^2 (1-Q^2) D_e}{F}}. \tag{17}$$

The microwave equation therefore contains $D_b + \Sigma_\perp$. The quantity $\Sigma_\perp$ is the response of the eliminated pair: its real part adds amplitude damping and its imaginary part changes the dispersive response. The associated additional full linewidth is $2\,\mathrm{Re}\,\Sigma_\perp$. Although this pair has no direct optical-cavity coupling, the microwave field drives it; it feeds back on that field and can dissipate through atomic decay. Simply replacing G_μ by QG_μ retains the useful projection but drops this feedback. Neither $\Sigma_\perp$ nor the baseline extra damping belongs in the useful output-port rate.

2.5 Closed transfer function.

After eliminating the orthogonal pair, put $B = D_b + \Sigma_\perp$. Eliminating the remaining atomic amplitudes gives

$$S_{mo} = \frac{\sqrt{\kappa_{o,\mathrm{ex}}\kappa_{m,\mathrm{ex}}}\, g_o g_m \Omega Q}{B(D_a F + g_o^2 D_r) + g_m^2 Q^2 (D_a D_e + g_o^2)}. \tag{18}$$

Define $Y = D_a F + g_o^2 D_r$ and $Z = D_b F + g_m^2 D_e$. Substituting for $\Sigma_\perp$ and using $F = D_e D_r + \Omega^2$ then gives the compact result, up to an overall output phase,

$$\boxed{S_{mo}(\omega) = \frac{\sqrt{\kappa_{o,\mathrm{ex}}\kappa_{m,\mathrm{ex}}}\, g_o g_m \Omega Q F}{YZ + Q^2 g_o^2 g_m^2 \Omega^2}, \qquad \eta(\omega) = |S_{mo}(\omega)|^2.} \tag{19}$$

Q changes both the numerator and the resonant response in the denominator. This is why the matched-profile efficiency cannot generally be scaled by Q^2. At $Q = 1$, the orthogonal pair decouples and the four-mode baseline model is recovered; at $Q = 0$, coherent conversion vanishes.

3 On-resonance efficiency and the numerical value of Q

At $\omega = \Delta_e = \Delta_r = \Delta_\mu = 0$, introduce the dimensionless cooperativities, which compare coherent coupling strengths with damping:

$$C_o = \frac{2G^2}{\kappa_o \Gamma}, \qquad C_m = \frac{2G_\mu^2}{(\kappa_m + \gamma_r)\gamma_r}, \qquad C_B = \frac{4\Omega^2}{\Gamma \gamma_r}. \tag{20}$$

Let $\eta_o = \kappa_{o,\mathrm{ex}}/\kappa_o$ and $\eta_m = \kappa_{m,\mathrm{ex}}/\kappa_m$. Equation (19) becomes

$$\boxed{\eta(0) = \eta_o\eta_m \frac{\kappa_m}{\kappa_m+\gamma_r} \frac{4Q^2C_oC_mC_B(1+C_B)^2}{\left[(1+C_B+C_o)(1+C_B+C_m)+Q^2C_oC_mC_B\right]^2}.} \tag{21}$$

The port fractions multiply this expression at fixed total linewidths. If a change in mirror or microwave-port coupling changes a total linewidth, that linewidth and the cooperativities must also be updated. Additional propagation or detection losses multiply the port-to-port result when defining an end-to-end efficiency.

3.1 Lattice example.

Use ^{85}Rb, $\lambda_\mathrm{L} = 1188\,\mathrm{nm}$, waist $w = 200\,\mu\mathrm{m}$, 100% retroreflection, and the specified calibration of 3 W per beam per axis to $U/k_\mathrm{B} = 15\,\mu\mathrm{K}$. This is an adopted depth calibration, not a new polarizability calculation. The recoil energy is $E_\mathrm{R}/k_\mathrm{B} = 0.0799004\,\mu\mathrm{K}$. For $T = 0$ and two equal, exactly matched second-order axes,

$$b_2 = e^{-4\sqrt{E_\mathrm{R}/U}} = 0.746816, \qquad \boxed{Q = b_2^2 = 0.557734}, \qquad Q^2 = 0.311067. \tag{22}$$

The $T = 0$ value is the harmonic lowest-band approximation and includes zero-point motion. It is a static overlap input to the model; motion-resolved sidebands are not calculated here.

3.2 Converter example.

For a reproducible reference case, take the following collective coupling and linewidth parameters:

$$\begin{aligned} \frac{(G, G_\mu, \Omega)}{2\pi} &= (15.8474565,\ 2.09170145,\ 5.52497339)\,\mathrm{MHz}, \\ \frac{(\kappa_o, \kappa_m, \Gamma, \gamma_r)}{2\pi} &= (10,\ 5,\ 6,\ 0.1)\,\mathrm{MHz}. \end{aligned} \tag{23}$$

All detunings are zero. Since the efficiency is dimensionless, one may evaluate it using the listed MHz numbers if every rate and frequency is divided by the same 2π factor. With $\eta_o = \eta_m = 1$, and Q from Eq. (22) for the two finite-depth cases:

Model	On-resonance efficiency
Matched profiles, $Q = 1$	82.0957%
Only $G_\mu \mapsto QG_\mu$; orthogonal pair omitted	51.5929%
Six modes with $Q = 0.557734$	**47.0824**%

Here $\Sigma_\perp(0)/(2\pi) = 0.147394\,\mathrm{MHz}$, corresponding to an additional full linewidth $2\,\mathrm{Re}\,\Sigma_\perp(0)/(2\pi) = 0.294787\,\mathrm{MHz}$. For example, desired-port fractions of 0.98 at both cavities give 45.21798% at unchanged total linewidths, before any further path losses.

3.3 Scope and checks.

The closure assumes weak excitation, a frozen spatial distribution, uniform control magnitude and atomic linewidths, and one effective optical mode profile. Anharmonic motion, resolved motional sidebands, differential light shifts, and further independent standing-wave channels require an extended susceptibility model. The closed expression was checked against the six-equation solution and the $Q = 1$ limit in Mathematica. The efficiencies above are conditional model predictions for the stated geometry.

3.4 Using the companion Mathematica notebook.

The companion `Abhishek_Converter_QPM.nb` follows the same progression: solve the six equations, eliminate the orthogonal pair, check the $Q = 1$ limit, and substitute numerical parameters. Its optical linewidth κ is κ_o here, and its microwave linewidth κ_μ is κ_m. The symbols G, G_μ, Ω, Γ, γ_r, and the detunings retain their definitions above. The calculator names G_{opt} and G_{mm} correspond to G and G_μ. Set the notebook's auxiliary parameter $\epsilon = 1$ when comparing with this note.

The notebook retains a forcing parameter `Ein` in its first equation and defines `ConverterEfficQ` as $\sqrt{\kappa_o \kappa_m}\, b$ in that normalization. It is therefore an amplitude, despite its name. The connection to the physical, input-normalized response is

$$S_{mo} = \sqrt{\eta_o \eta_m}\, \frac{\texttt{ConverterEfficQ}}{\texttt{Ein}}, \qquad \eta = \eta_o \eta_m\, \texttt{NormalizedEfficiencyQ}, \tag{24}$$

up to the same overall port phase, with $\epsilon = 1$. Here `NormalizedEfficiencyQ` is `EfficiencyQ`/`Ein`^{2} for the real input convention in the notebook. The numerical examples use `Ein`=1; a different drive amplitude must not change the normalized efficiency in this linear model.

The stored expression `SigmaPerp` is Eq. (17), and `OnResonanceEfficiencyQ` is Eq. (21) with unit port fractions. The preset `parmsScreenshot` contains the reference parameters of Eq. (23); `parmsOriginal` is a different parameter set from the original notebook. Keep Q symbolic until applying `qRule` in the numerical example. The notebook calls its two matched axes x, z instead of x, y; these are labels for the same two-factor overlap assumption, not an extra axis. Finally, `etaOpt` and `etaMM` are the port fractions η_o and η_m, applied through `EfficiencyQWithPorts` at fixed total linewidths.

References

[1] I. Bloch, J. Dalibard and W. Zwerger, *Rev. Mod. Phys.* **80**, 885 (2008), Sec. II. doi:10.1103/RevModPhys.80.885.
[2] A. Kumar et al., *Nature* **615**, 614–619 (2023). doi:10.1038/s41586-023-05740-2.
[3] Baseline model: `Abhishek_Converter.nb`; companion implementation: `Abhishek_Converter_QPM.nb`. Calculator: AVKGitHub1/transduction_calcs, commit `30da8841b3c9`. The Q extension is derived here.